



JULIAN FRITSCH

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Nationality: German

Swiss Residence Status: C permit

AREAS OF EXPERTISE

Applied Artificial Intelligence, Production ML Engineering, Biomedical Signal Analysis, NLP & LLM

WORK EXPERIENCE

Senior IT Engineer

Groupe Mutuel

July 2023 – now

Sion, Switzerland

- Engineered full-stack AI/ML process automation contributing to team efficiency savings of 40 FTEs.
- Integrated ML text-processing pipelines for 1M+ annual inferences (~10k/day) into internal workflows.
- Architected and deployed 2 production LLM extraction pipelines (LangChain, Pydantic), automatically processing 200k+ documents per year.

Applied AI Engineer (Part-time)

Virtuosis AI

Apr 2026 – Jun 2026

Lausanne, Switzerland

- Deep learning models for vocal biomarker extraction and health state prediction
- Presented the company at the VVG MediCongress 2026.

Research Intern

Fraunhofer Audio Labs

Nov 2016 – Mar 2017

Erlangen, Germany

- Applying neural networks to audio bandwidth extension

Research Intern

Eriksholm Research Center (Oticon)

Nov 2015 – Mar 2016

Helsingør, Denmark

- Implementing a hearing aid acoustic feedback elimination algorithm in Assembler

EDUCATION

Docteur es Sciences | *Novel Methods For Detection And Analysis Of Atypical Aspects In Speech* 2018 – 2023

École polytechnique fédérale de Lausanne (EPFL) & Idiap Research Institute

Switzerland

Master of Science | *Medical image and data processing*

2015 – 2018

Friedrich Alexander Universität (FAU) Erlangen-Nuremberg

Germany

Bachelor of Arts | *Medical Engineering: Biomedical signal analysis, pattern recognition*

2010 – 2015

Friedrich Alexander Universität (FAU) Erlangen-Nuremberg

Germany

TECHNICAL SKILLS

Languages: Python (async application development), Bash, SQL

AI, LLM & Agentic Ecosystem: LangChain, Pydantic, RAG

Developer Tools: Git, Vim, VS Code

Observability: Splunk, Grafana

Libraries: Pandas, LangChain, Pydantic, Huggingface Transformers

Deep Learning Libraries: Keras, PyTorch

Architecture: FastAPI, Apache Kafka, Docker, CI/CD

LANGUAGES

German: Native language

English: Full professional proficiency

French: Full professional proficiency

PUBLICATIONS: [SCHOLAR.GOOGLE.COM/CITATIONS?USER=LJMXDZCAAAAJ](https://scholar.google.com/citations?user=LJMXDZCAAAAJ)